DIMENSION 01 -> Responsiveness & Support Speed

Hypothesis test 01 -> numeric variable vs retention label

response minutes vs retention label

1) Defining the hypothesis

- **Null hypothesis :** The distribution of response_minutes is the same for retained and non-retained customers.
- **Alternate hypothesis:** The distribution of response_minutes is different for retained and non-retained customers.

What is the dependent and independent variable and why is it important for this hypothesis test?

For this test:

- **Independent variable:** retention_label
- **Dependent variable:** response_minutes

Why?

Because in this hypothesis test, retention_label is used to **form the two groups**:

- retained customers
- non-retained customers

Then we examine whether the **measured outcome**, response_minutes, differs between those groups.

So:

- retention_label = grouping / explanatory variable
- response_minutes = outcome / measured variable

Why is this important?

It is important because it tells us **what is being compared and how**. In this test, we are asking: Does the value of response_minutes change depending on whether the customer is retained or not? That matters because the test setup depends on it:

- one variable must define the groups
- the other variable must provide the numeric values being compared across those groups

In simple words

- **Independent variable** answers: "Which group does this row belong to?"
- **Dependent variable** answers: "What numeric value are we comparing between groups?"

Here:

- group = retention_label
- compared numeric values = response_minutes

what does distribution mean in the hypothesis given?

Here, **distribution** means how the response_minutes values are spread within each group:

- where the values tend to lie
- whether they are mostly small or large
- the overall pattern of the values

If you go trough the datasets, you can clearly see how the response_minutes values are in the rows. So it means comparing:

- the set of response_minutes values for retention_label = 1
- with the set of response_minutes values for retention_label = 0

If the numeric variable is considered as response_minutes then why do we consider data points to be retention label?

Because the test is asking: **Does response_minutes differ between retained and non-retained customers?**

So:

- **response_minutes** = the numeric variable being analyzed
- **retention_label** = the grouping variable that splits the data into 2 groups:
    - retained (1)
    - not retained (0)

We are **not treating retention_label as the numeric measurement**.

We are using it only to **separate the rows into groups** and then compare the response_minutes values between those groups. Technically we are comparing the distribution of the response_minutes values in both the group retained and not retained.

2) Test statistic and its distribution

For numerical (continuous or quantitative) features, hypothesis tests are chosen based on the **number of groups being compared**, the **independence of samples**, and the **distribution of the data** (normal vs. non-normal). For a numeric variable there are different types of tests we can perform, and they are listed as follows,

Scenario	Data Type	Test to Use
1 Sample Mean	Normal	One-sample T-test
2 Independent Means	Normal	Independent T-test
2 Independent Groups	Non-normal	Mann-Whitney U Test
Paired Samples	Normal	Paired T-test
Paired Samples	Non-normal	Wilcoxon Signed-Rank Test
3+ Independent Groups	Normal	One-way ANOVA
3+ Independent Groups	Non-normal	Kruskal-Wallis Test
Relationship	Continuous	Pearson Correlation

x ✓ Parametric non-parametric

Based on the selection categories listed above and hypothesis test choosing criteria we do a justification. This justification will conclude as to why we selected the relevant hypothesis test and distribution. From this we will conclude that we will be using Mann-Whitney U test BUT what I have circled in black is what we learned under lectures. We use 3 layer decision strategy,

layer 1) number of groups being compared,

What we check

We first check how many groups the outcome is being compared across.

Here:

- response_minutes = numeric variable
- retention_label = grouping variable

From the dataset, retention_label has **2 groups**:

- 0 = non-retained = 2278 rows
- 1 = retained = 420 rows

So the comparison is between **two groups only**.

Why this matters : The number of groups determines the **family of tests: (refer to the table of tests I have given above)**

- **2 independent groups** → possible tests:
  - Independent t-test / Welch t-test
  - Mann–Whitney U test
- **3 or more groups** → possible tests:
  - ANOVA
  - Kruskal–Wallis

Since we have **only 2 groups**, we do **not** use ANOVA or Kruskal–Wallis. So, the decision reduces to: Should we consider independent or dependent groups ?

layer 2) independence of samples

What we check -> We must decide whether the two samples are **independent** or **paired/dependent**.
Independent samples mean:
A value in one group does not have a natural one-to-one paired partner in the other group.
Examples of **paired/dependent** data:

- before vs after measurements on the same person
- left hand vs right hand of same person
- same customer measured twice

Examples of **independent** data:

- one group of retained customers vs a different group of non-retained customers

In this dataset
Each row is treated as **one observation / one service case**.
A row belongs to exactly one group:

$$retention\_label \in \{0,1\}$$

So each response_minutes value is in either:

- retained group
- non-retained group

but not both.
That means the groups are **mutually exclusive**.

Conceptually:

$$X_1 = \{x_{11}, x_{12}, \ldots, x_{1n_1}\}$$
$$X_0 = \{x_{01}, x_{02}, \ldots, x_{0n_0}\}$$

where:

- X_1 = response times of retained customers
- X_0 = response times of non-retained customers

There is **no matching pair structure** like $(x_{1i}, x_{0i})$for the same subject.

Independence is partly a **study-design assumption**, not something we can fully prove from the CSV alone. Because our CSV file has no customer ID / repeated-measure flag, we **assume** rows are separate observations.
If the same customer appears many times, strict independence could be weakened.

So the correct statement is: Based on the available dataset structure, the samples are treated as independent, **because observations are split into two distinct groups with no explicit pairing variable.**

Why this matters
This determines whether we use:

- **paired tests**:
  - paired t-test
  - Wilcoxon signed-rank test
- **independent-sample tests**:
  - independent t-test / Welch t-test
  - Mann–Whitney U test

Since the samples are treated as **independent**, paired tests are not appropriate. Therefor now selection strategy concludes down to **independent 2** sample groups. we can now either use parametric independent t-test / Welch t-test or Mann–Whitney U test, next part confirms this.

layer 3) Distribution of data (normality vs non-normality)
We consider parametric and non-parametric nature and it is defined as follows. We need to conclude the test statistics and

parametric or non- parametric nature helps us to do this. Immediately we can tick some observable features such as **data type, sample** size but some features such as **distribution, robustness, central tendency** need a careful analysis. The once we need as careful analyses are being checked to determine whether to go with t-test which is what sir taught us or to go with Mann-Whitney.

Feature	Parametric	Non-Parametric
Distribution	Assumes normal/known	No assumption ("free")
Data Type	Continuous	Ordinal/Ranked/Nominal
Central Tendency	Mean	Median
Sample Size	Large	Small to Large
Robustness	Sensitive to outliers	Robust to outliers
Examples	? t-test, ANOVA	? Mann-Whitney, Wilcoxon

Normality, outliers and central tendency check -> What we check

For a numeric variable like response_minutes, we examine whether the variable is approximately normal **within each group**.

That means checking normality separately for:

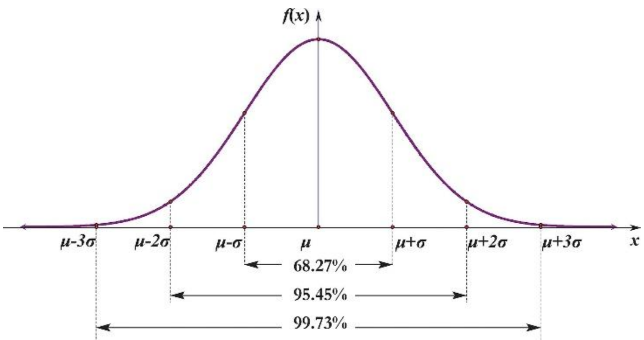
- retained group
- non-retained group

This matters because:

- if approximately normal → t-test / Welch t-test may be suitable
- if strongly non-normal → Mann–Whitney U is safer.

Theory 01 -> What does normality and non-normality mean?

Normality refers to data that follows a symmetrical, bell-shaped Gaussian distribution, where the mean, median, and mode are equal, with roughly 68%, 95% and 99% data falling within 1, 2, and 3 standard deviations. Non-normality indicates data deviates from this shape, appearing skewed or multimodal, often requiring non-parametric tests or data transformation. Normality is a probability distribution that is symmetric about the mean. Non-normality is data that does not follow the Gaussian bell-shaped pattern, often skewed to the left or right, or having multiple peaks (multimodal).



Theory 02 -> Then why do we need to consider normality for t-test and non-normality for Mann-Whitney?

T-test relies on the assumption that data follows a normal distribution (bell curve) to validly calculate means and standard deviations. **Sensitivity to Skew/Outliers:** If data is highly skewed or contains severe outliers, the mean becomes an unreliable measure of central tendency, making test results misleading. **Sample Size Impact:** While t-tests are somewhat robust to mild violations with larger sample sizes (due to the Central Limit Theorem), extreme non-normality invalidates them, particularly with smaller samples.

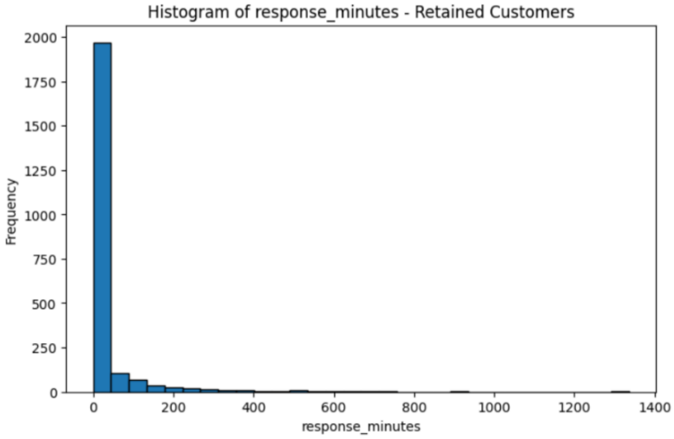
The Mann-Whitney U test does not assume normality, making it ideal for skewed data, ordinal data (likert scales), or data with extreme outliers. **Rank-Based Analysis:** Instead of raw values, this test uses the ranks of the data points, which diminishes the impact of extreme outliers or non-normal distribution shapes. **Independent Groups:** It is designed to compare two independent, non-normally distributed samples.

Evidence 01 -> Measuring normality using visual/descriptive data

We need to consider both visual data and descriptive data to first gather evidence information needed to check for the normality. We derive a **histogram plot and descriptive data** from the dataset as follows.

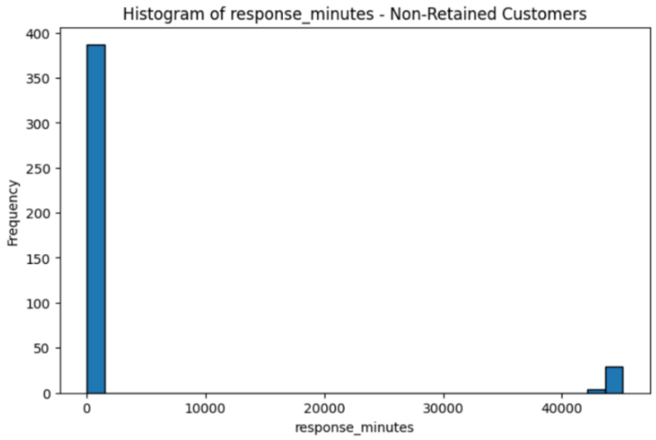
For retained customers:

- mean = 30.50
- median = 4
- std = 86.35
- skewness = 5.75
- kurtosis = 46.71



For non-retained customers:

- mean = 3538.80
- median = 16
- std = 11900.17
- skewness = 3.13
- kurtosis = 7.82



If data were close to normal:

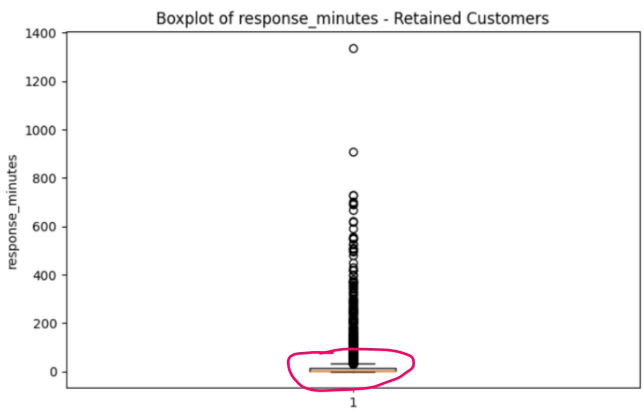
- mean and median should be relatively close, skewness should be near 0, kurtosis should not be extremely large

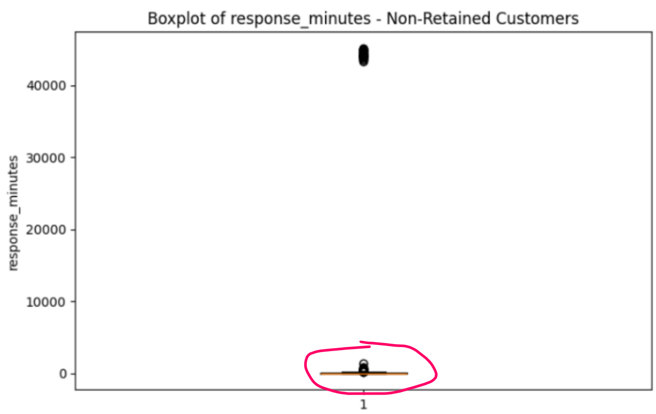
But here:

- means are far from medians, skewness is very high, kurtosis is extremely high, there are large outliers / long right tails.

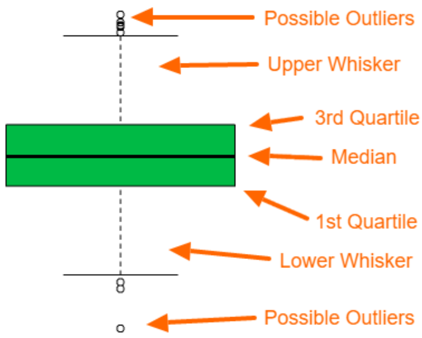
A histogram confirms non-normality by revealing a data distribution that lacks the characteristic symmetrical, single-peaked "bell curve" shape, typically showing skewness, multiple peaks, or heavy tails. It acts as a visual tool to identify departures from symmetry, where the left and right sides of the centre do not mirror each other.

This is strong evidence of non-normality. (apply theory 01)





The above boxplots also confirm non normality as,



- **Asymmetry/Skewness:** The median line is not cantered within the box. If it is closer to the bottom, the data is positively (right) skewed; if closer to the top, it is negatively (left) skewed.
- **Unequal Whiskers:** One whisker is significantly longer than the other, indicating a longer tail in that direction.
- **Outliers:** Individual dots (or asterisks) appearing outside the whiskers indicate extreme values, violating the expectations of a normal distribution.
- **Box Proportion:** In a normal distribution, the median line is roughly in the middle, and the quartiles (edges of the box) are symmetric

Boxplots confirm non-normality. This is strong evidence of non-normality. (apply theory 01)

Evidence 02 -> Measuring normality using Shapiro–Wilk test

The Shapiro-Wilk test is a statistical test used to determine whether a dataset follows a normal (Gaussian) distribution.

Hypotheses:

- **H_0:** data are normally distributed
- **H_1:** data are not normally distributed

P value is being obtained using the following code

```
from scipy.stats import shapiro

print(shapiro(g1.sample(500, random_state=0)))
print(shapiro(g0.sample(min(500, len(g0)), random_state=0)))
```

Actual results

- retained group: **$p \approx 1.37 \times 10^{-38}$**
- non-retained group: **$p \approx 6.69 \times 10^{-37}$**

Since both p-values are far below 0.05: (we consider significance level 0.05)

$$p < 0.05 \Rightarrow \text{reject normality}$$

Both groups are not normally distributed.

Evidence + theory -> concluding the Normality, outliers and central tendency

Based on the theory and evidence now we can make a conclusion about the normality, outliers and central tendency. Shapiro wilk test uses a significance level of 0.05, this can cause a potential issue from type 1 error (rejecting null hypothesis when it is correct). Therefore, for further confirmation we use histogram + descriptive strategy. Both of this confirmed that the data in both groups follows a non-normality. From theory 01 we described; we can conclude evidence 01 will show a non-normality.

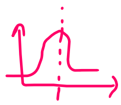
Under non-normality as given in theory 02, **t-test** would result in wrong results for the hypothesis due to extreme skewness and severe outliers. The t-test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}}$$

This is appropriate mainly when: samples are independent, outcome is numeric, distributions are reasonably normal, or the mean is the right summary measure.

A normal distribution is symmetric and bell-shaped, with density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$



response_minutes values do not behave like that: many observations are small, a few are extremely large, distribution is strongly right-skewed. That creates: large skewness, large kurtosis, means pulled upward by outliers

But **Mann-Whitney u-test** is suitable under the given conditions mentioned in theory 02 and it wins for the situations large skewness, large kurtosis, means pulled upward by outliers. For large samples, under H_0, the distribution of U is approximated by a normal distribution (this does not mean data should follow normality):

$$U \sim N(\mu_U, \sigma_U^2)$$

where

$$\mu_U = \frac{n_1 n_2}{2}$$

$$\sigma_U = \sqrt{\frac{n_1 n_2 (n_1 + n_2 + 1)}{12}}$$

In your data:

- $n_1 = 2278$
- $n_2 = 420$

Since both are large, the **normal approximation to U** is appropriate.

Therefor we can conclude that Mann-Whitney u-test is the best.

Feature	Parametric	Non-Parametric
Distribution	Assumes normal/known	No assumption ("free")
Data Type	Continuous	Ordinal/Ranked/Nominal
Central Tendency	Mean	Median
Sample Size	Large	Small to Large
Robustness	Sensitive to outliers	Robust to outliers
Examples	t-test, ANOVA	Mann-Whitney, Wilcoxon

The Mann-Whitney U test (or Wilcoxon rank-sum test) is a non-parametric, independent two-sample test used to determine if there is a significant difference between the distributions of two groups. It serves as an alternative to the t-test when data is ordinal or not normally distributed, comparing median ranks rather than means. . Less affected by outliers than parametric tests because it relies on ranks rather than raw data.

Refer to the example done on how to do Mann Whitney test mathematically.

Example:

You measured the reaction time of a small group of men and women and want to know if there is a difference.

Gender	Reaction time	Rang
female	34	2
female	36	4
female	41	7
female	43	9
female	44	10
female	37	5
male	45	11
male	33	1
male	35	3
male	39	6
male	42	8

Calculation of the rank sums

$$T_1 = 2 + 4 + 7 + 9 + 10 + 5 = 37$$

$$T_2 = 11 + 1 + 3 + 6 + 8 = 29$$

Data are not normally distributed → Mann-Whitney U Test

Null hypothesis

Both rank sums are the same.

Female

Number of cases $n_1 = 6$ Rank sum $T_1 = 37$

$$U_1 = n_1 \cdot n_2 + \frac{n_1 \cdot (n_1 + 1)}{2} - T_1 = 6 \cdot 5 + \frac{6 \cdot (6 + 1)}{2} - 37 = 14$$

Male

Number of cases $n_2 = 5$ Rank sum $T_2 = 29$

$$U_2 = n_1 \cdot n_2 + \frac{n_2 \cdot (n_2 + 1)}{2} - T_2 = 6 \cdot 5 + \frac{5 \cdot (5 + 1)}{2} - 29 = 16$$

U-Wert

$$U = min(U_1, U_2) = min(14, 16) = 14$$

Expected value of U

$$\mu_U = \frac{n_1 \cdot n_2}{2} = \frac{6 \cdot 5}{2} = 15$$

Standard error of U

$$\sigma_U = \sqrt{\frac{n_1 \cdot n_2 \cdot (n_1 + n_2 + 1)}{12}} = \sqrt{\frac{6 \cdot 5 \cdot (6 + 5 + 1)}{12}} = 5.4772$$

z-value

$$z = \frac{U - \mu_U}{\sigma_U} = \frac{14 - 15}{5.4772} = -0,1825$$

3) Computing the test statistic under the null hypothesis

```python
# keep only needed columns and remove missing values
sub = df[["response_minutes", "retention_label"]].dropna().copy()
sub["retention_label"] = sub["retention_label"].astype(int)
# split into 2 groups
retained = sub[sub["retention_label"] == 1]["response_minutes"]
not_retained = sub[sub["retention_label"] == 0]["response_minutes"]

# Mann-Whitney U test
result = mannwhitneyu(retained, not_retained, alternative="two-sided")
U = result.statistic
p_value = result.pvalue

# values under H0
n1 = len(retained)
n2 = len(not_retained)

mu_U = (n1 * n2) / 2
sigma_U = math.sqrt((n1 * n2 * (n1 + n2 + 1)) / 12)

# standardized Z value
Z = (U - mu_U) / sigma_U
```

Z-value = -10.64
U = 322260,
p-value = 1.22×10^{-26},

4) Defining the rejection region

Using significance level:

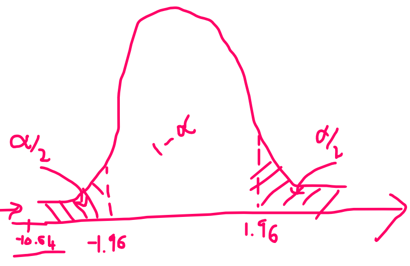
$$\alpha = 0.05$$

For a two-tailed test, reject H_0 if:

$$|\,Z\,| > 1.96$$

So, the rejection region is:

$$\boldsymbol{Z < -1.96 \text{ or } Z > 1.96}$$



Observed value:

$$\boldsymbol{Z = -10.64}$$

Since $-10.64 < -1.96$, the test statistic falls in the rejection region.
Also:

$$p = 1.22 \times 10^{-26} < 0.05$$

So, we reject H_0.

5) CONCLUSION

There is a **statistically significant difference** in response_minutes between retained and non-retained customers. So, response_minutes has a significant relationship with customer retention in this dataset.

In practical terms:

- retained customers had much lower response times
- non-retained customers had much higher response times

This suggests that **faster response time is associated with better customer retention**.

Hypothesis test 02 -> categorical variable vs retention label

category vs retention_label

1) Defining the hypothesis

Both variables are categorical:

- category = complaint / service category
- retention_label = whether the customer was retained (1) or not retained (0)

So, we test whether these two variables are **independent** or **associated**.

- **Null hypothesis (H_0):** category and retention_label are independent.
- **Alternative hypothesis (H_1):** category and retention_label are associated.

what is the dependent and independent variable and why is it important for this hypothesis test?

For **category vs retention_label**:

- **Independent variable:** category
- **Dependent variable:** retention_label

Why?

Because the question is basically: Does the **category** of the case relate to whether the customer is **retained or not**?

So:

- category acts as the explanatory / grouping variable
- retention_label is the outcome variable being examined across those categories

Why is this important for this hypothesis test?

It is important because it clarifies the interpretation:

- we are testing whether **retention outcome changes across categories**, not whether category is “caused” by retention

Note that there is a difference between how we chose the dependent and independent variable under hypothesis test 1 and hypothesis test 2 (current test). Why?

For hypothesis 02, **category vs retention_label**:

- **Independent variable:** category
- **Dependent variable:** retention_label

For hypothesis 01 **response_time vs retention_label:**

- **Independent variable:** retention_label
- **Dependent variable:** response_minutes

Why? Lets see!!

Hypothesis 01: response_minutes vs retention_label

Using **Mann-Whitney U**:

- retention_label is the **grouping variable**
- response_minutes is the **measured numeric variable**

So **inside that test setup**:

- Independent variable = retention_label
- Dependent variable = response_minutes

Because the test literally asks:

Are the response_minutes values different **between the two retention groups**? So in the test mechanics, retention defines the groups, and response time is what is compared. Any changes to group values can alter the test result and distribution of response_minutes theoretically.

Hypothesis 02: category vs retention_label

Using **Chi-square test of independence**:

Both variables are categorical, and the chi-square test is actually **symmetric**.

That means mathematically:

- category vs retention_label
- retention_label vs category

give the **same chi-square statistic and same p-value**. So here, calling one "independent" and the other "dependent" is mainly for **interpretation**.
We chose:

- Independent variable = category
- Dependent variable = retention_label

because the business meaning is: Does retention outcome vary across categories? So here retention_label is treated as the outcome of interest.

2) Test statistic and its distribution

For this we will be using a method we have been taught in the lecture.

Because:

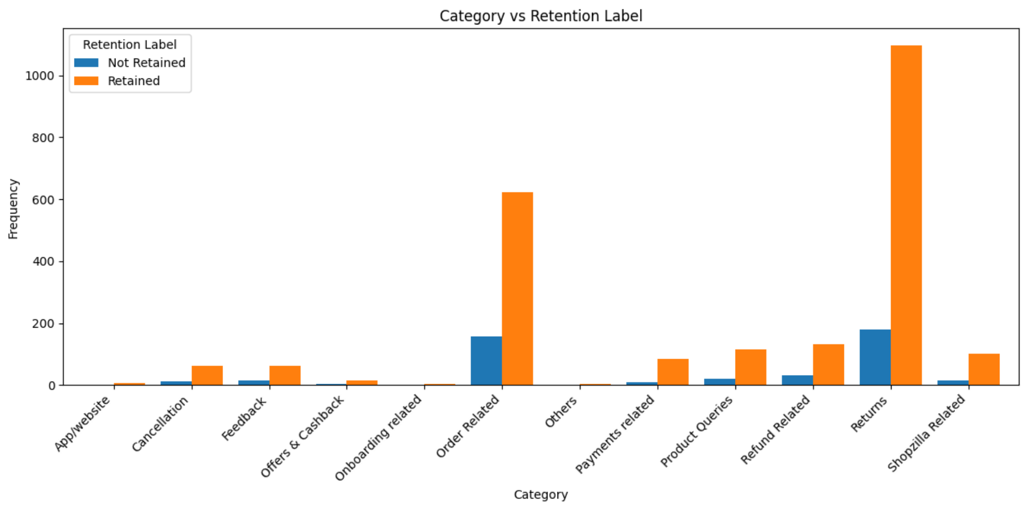
- we are comparing **two categorical variables**
- the data are arranged as counts in a contingency table (given below check it)

the appropriate test is the **Chi-square test of independence**. This is the standard, most widely used test for two categorical variables, particularly for larger sample sizes. To determine if there is a significant association between two categorical variables (e.g., gender and political preference). **Assumptions:** Independent observations and a large enough sample size (expected frequencies in each cell of the contingency table should be greater than 5).

Why this test is suitable

- both variables are categorical
- the data are frequency counts
- we want to test **association / independence**, not compare means or medians

By visualising the retention label and category we can further confirm the use of chi squared test.



Test statistic
The test statistic is:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where:

- O_{ij} = observed frequency in cell i, j
- E_{ij} = expected frequency in cell i, j under H_0

Expected frequencies under H_0

$$E_{ij} = \frac{(\textbf{row total}_i)(\textbf{column total}_j)}{N}$$

Distribution under H_0

Under the null hypothesis, the test statistic approximately follows a **Chi-square distribution**:

$$\chi^2 \sim \chi^2_{(r-1)(c-1)}$$

where:

- r= number of rows
- c= number of columns

From your dataset:

- number of category levels = **12**
- number of retention_label levels = **2**

So: degree of freedom(df);

$$df = (12-1)(2-1) = 11$$

Therefore, under H_0:

$$\chi^2 \sim \chi^2_{11}$$

3) Computing the test statistic under the null hypothesis

The following code derives the chi squared value and the p-value.

```
file_name = "dimension_01_stage_01_preprocessing_part02.csv"
df = pd.read_csv(file_name)
d = df.dropna(subset=["category", "retention_label"]).copy()
d["retention_label"] = d["retention_label"].astype(int)

table = pd.crosstab(d["category"], d["retention_label"])
chi2_stat, p_value, dof, expected = chi2_contingency(table)

print(table)
print("Chi-square:", chi2_stat)
print("p-value:", p_value)
print("df:", dof)
```

```
retention_label      0     1
category
App/website          0     5
Cancellation        13    61
Feedback            15    61
Offers & Cashback    2    14
Onboarding related   0     3
Order Related      156   624
Others               0     3
Payments related     8    84
Product Queries     21   114
Refund Related      30   133
Returns            178  1097
Shopzilla Related   15   100
Chi-square: 21.51480064762059
p-value: 0.028413380700258895
df: 11
```

Mathematically this is how we derive the chi squared value;

Observed contingency table

Category	**Retention=0**	**Retention=1**
App/website	0	5
Cancellation	13	61
Feedback	15	61
Offers & Cashback	2	14
Onboarding related	0	3
Order Related	156	624
Others	0	3

Payments related	8	84
Product Queries	21	114
Refund Related	30	133
Returns	178	1097
Shopzilla Related	15	100

Totals:

- total $N = 2737$
- column totals:
  - retained = 2299
  - not retained = 438

Expected frequencies under H_0

Under independence:

$$E_{ij} = \frac{(\text{row total}_i)(\text{column total}_j)}{N}$$

Example for **Cancellation, retention = 0**:

- row total for Cancellation = 74
- column total for retention = 0 is 438
- grand total = 2737

$$E = \frac{74 \times 438}{2737} \approx \mathbf{11.84}$$

Example for **Cancellation, retention = 1**:

$$E = \frac{74 \times 2299}{2737} \approx \mathbf{62.16}$$

Then each cell contributes:

$$\frac{(O - E)^2}{E}$$

For Cancellation:

- retention = 0:

$$\frac{(13 - \mathbf{11.84})^2}{11.84} \approx 0.11$$

- retention = 1:

$$\frac{(61 - \mathbf{62.16})^2}{62.16} \approx 0.02$$

We do this for **all 24 cells** and sum them. We have only done for 2 cells (cancellation retention label=0 and cancellation retention label =1)

Final computed statistic

From the dataset:

$$\chi^2 = 21.5148$$
$$p = 0.0284$$
$$df = 11$$

4) Defining the rejection region

Take significance level:

$$\alpha = 0.05$$

For a Chi-square test, reject H_0 if the observed test statistic is greater than the critical value:

$$\chi^2_{\text{observed}} > \chi^2_{0.95,11}$$

The critical value is:

$$\chi^2_{0.95,11} \approx 19.675$$

So the rejection region is:

$$\chi^2 > 19.675$$

Observed value:

$$\chi^2 = 21.5148$$

Since:

$$21.5148 > 19.675$$

the test statistic falls in the rejection region.
Equivalently, using the p-value rule:

$$p = 0.0284 < 0.05$$

So we reject H_0.

5) Conclusion

There is a **statistically significant association** between category and retention_label at the 5% significance level.
So we conclude that: Customer retention is not independent of category in this dataset.
In simple words:

- the distribution of retained vs non-retained customers changes across different categories
- some categories are associated with relatively better or worse retention outcomes

Hypothesis test 3.1 -> numeric variable vs retention label (categorical)

response minutes vs retention label

Now we will see how linear regression models the relationship between response_minutes and retention label.

Using **linear regression** for this hypothesis, we set:

$$\text{response\_minutes}_i = \beta_0 + \beta_1(\text{retention\_label}_i) + \varepsilon_i$$

where:

- response_minutes = dependent variable
- retention_label = independent variable (0 = not retained, 1 = retained)

This is a simple linear regression with a **binary predictor**.

Why do we select dependent variable as response_minutes and independent variable retention_label?

Identify the Goal (Dependent Variable - Y) -> The variable that is being predicted, measured, or affected.

Identify the Predictor (Independent Variable - X) -> The variable used to explain or cause changes in the dependent variable.

According to above definitions, what we have selected as dependent variable and independent variable is based on the business logic or the hypothesis logic.

Usually, we can't use linear regression to model direct relationship between numerical variable(response time) and categorical variable (retention label). As explained under Note-10 linear regression, there are assumptions to consider in linear regression, is the choice taken by considering them?

from the assumptions,

Linearity → acceptable

Multicollinearity → not relevant

Homoscedasticity → violated

Independence → assumed, not fully verifiable

Normality → violated

Our hypothesis is based on this linearity factor, what does linearity say? Our lecture note gives a description like, Linearity: The relationship between X and the mean of Y is linear. This is exactly what we consider for our hypothesis test.

predictor retention_label has only 2 values:

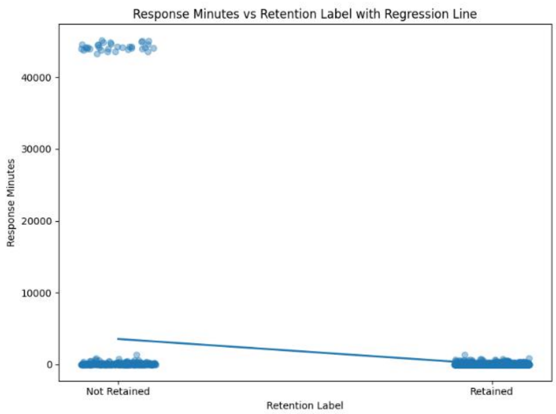
- 0 = not retained
- 1 = retained

With a binary X, the regression line just connects the two group means:

$$E(Y \mid X = 0) = \beta_0$$
$$E(Y \mid X = 1) = \beta_0 + \beta_1$$

So the "linearity" assumption is not difficult here. The model is simply saying: the mean response time for group 0 and the mean response time for group 1 can be represented by a straight line across $X = 0$ and $X = 1$. Since there are only two X-points, that is automatically fine.

For binary predictor + continuous outcome, linearity is not the main problem here.



1) Defining the hypothesis

We want to test whether retention_label has a significant linear effect on response_minutes. So the coefficient of interest is β_1.

Null hypothesis H_0**:** under H_0, the mean response_minutes is the same for retained and non-retained customers

$$\beta_1 = 0$$

Alternative hypothesis H_1**:** under H_1, the mean response_minutes is different for retained and non-retained.

$$\beta_1 \neq 0$$

Since **retention_label is binary**, this regression is equivalent to testing whether the **two group means** are different.

2) Test statistic and its distribution

In simple linear regression, to test whether the predictor is significant, we use the **t-statistic** for the slope coefficient:

$$t = \frac{\hat{\beta}_1 - 0}{SE(\hat{\beta}_1)}$$

Under H_0, this statistic follows a **t-distribution** with:

$$df = n - 2$$

because one intercept and one slope are estimated.
From our data we have rows up to : $n = 2698$, therefore degrees of freedom,

$$df = 2698 - 2 = 2696$$

So under the null hypothesis:

$$t \sim t_{2696}$$

Why do we consider t-test strategy here and any reason for selection of the equation based on the slope $(\hat{\beta}_1)$?
A t-test is used because the hypothesis concerns whether the regression slope coefficient is significantly different from zero. The slope $\beta_1$represents the linear effect of the predictor on the response, so it is the relevant parameter for testing predictor significance. The test statistic is $t = \hat{\beta}_1/SE(\hat{\beta}_1)$, where the standard error of the slope is calculated from the residual variation and the spread of the predictor values:

$$SE(\hat{\beta}_1) = \sqrt{\frac{SSE/(n-2)}{\sum(x_i - \bar{x})^2}}$$

with

$$SSE = \sum(y_i - \hat{y}_i)^2$$

Under the null hypothesis $H_0 \colon \beta_1 = 0$, this statistic follows a t-distribution with $n - 2$ degrees of freedom.

3) Computing the test statistic under the null hypothesis

First I have discussed the mathematical approach and then later how to use the code to ontain the test statistic under null hypothesis,

For **simple linear regression**,

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

the estimates are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} \quad 1$$

and

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where:

- $\bar{x}$= mean of all $x$values
- $\bar{y}$= mean of all $y$values

What they mean

- $\hat{\beta}_1$= slope
- $\hat{\beta}_0$= intercept

In our case,

If:

- y =response_minutes
- x =retention_label

then the model is:

$$\text{response\_minutes} = \beta_0 + \beta_1(\text{retention\_label}) + \varepsilon$$

Since retention_label is binary:

- $x = 0$ for not retained
- $x = 1$ for retained

So here:

$$\hat{\beta}_0 = \text{mean response time for } (x = 0$$
$$\hat{\beta}_1 = \text{mean response time for } (x = 1) - \text{mean response time for } (x = 0)$$

So with a binary predictor, the slope is just the **difference between the two group means**. We cannot use the equation labelled as 1 (shown above) to determine the beta values $\hat{\beta}_0$ and $\hat{\beta}_1$. Therefore, we can obtain relevant mean values from the dataset and then calculate the beta values.

From the fitted regression: (as I have mentioned, we got them means)

$$\hat{\beta}_0 = 3538.7976$$
$$\hat{\beta}_1 = -3508.2954$$

So the fitted model is:

$$\widehat{\text{response\_minutes}} = 3538.7976 - 3508.2954(\text{retention\_label})$$

This means:

- if retention_label = 0:

$$\hat{y} = 3538.7976$$

- if retention_label = 1:

$$\hat{y} = 3538.7976 - 3508.2954 = 30.5022$$

These match the group means:

- not retained mean = **3538.80**
- retained mean = **30.50**

Now compute the t-statistic:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$$

From the model:

- $SE(\hat{\beta}_1) = 249.1619$

So:

$$t = \frac{-3508.2954}{249.1619} \approx -14.0804$$

The p-value is:

$$p \approx 1.68 \times 10^{-43}$$

Now using the code we can obtain the same result,

```python
# Load data
df = pd.read_csv("dimension_01_stage_01_preprocessing_part02.csv")
data = df[["response_minutes", "retention_label"]].dropna().copy()
data["retention_label"] = data["retention_label"].astype(int)

# Define X and y
X = data["retention_label"]
y = data["response_minutes"]

# Add intercept
X = sm.add_constant(X)
# Fit linear regression
model = sm.OLS(y, X).fit()
# Estimated slope coefficient beta1
beta1_hat = model.params["retention_label"]
# Standard error of beta1
se_beta1 = model.bse["retention_label"]
# Compute t statistic under H0: beta1 = 0
t_stat = (beta1_hat - 0) / se_beta1
# p-value
p_value = model.pvalues["retention_label"]

print("Estimated beta1:", beta1_hat)
print("Standard Error:", se_beta1)
print("t statistic:", t_stat)
print("p-value:", p_value)
```

```
Estimated beta1: -3508.295424139804
Standard Error: 249.16190737171578
t statistic: -14.080384361907708
```

3) Computing the test statistic under the null hypothesis

take significance level:

$$\alpha = 0.05$$

For a **two-tailed t-test**, reject H_0if:

$$\mid t \mid > t_{0.975,2696}$$

Critical value:

$$t_{0.975,2696} \approx 1.9608$$

So the rejection region is:

$$\boldsymbol{t < -1.9608 \text{ or } t > 1.9608}$$

Observed test statistic:

$$t = -14.0804$$

Since:

$$-14.0804 < -1.9608$$

the statistic falls in the rejection region.
Also, using the p-value rule:

$$p = 1.68 \times 10^{-43} < 0.05$$

So, we reject H_0.

5) Conclusion

There is a **statistically significant linear relationship** between retention_label and response_minutes.
So we conclude that: response_minutes differs significantly between retained and non-retained customers.
Because the slope is negative:

$$\boldsymbol{\hat{\beta}_1 = -3508.2954}$$

moving from retention_label = 0 to retention_label = 1 is associated with a large **decrease** in mean response_minutes.

Hypothesis test 3.2 -> categorical variable vs retention label

category vs retention label

Now we will model the relationship between category and retention label. Here both variables we consider are categorical.

1) Defining the hypothesis

For **category vs retention_label**, the clean regression-based way is **binary logistic regression**:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + \cdots + \beta_{11} D_{11i}$$

where:

- $p_i = P(\text{retention\_label} = 1 \mid \text{category})$
- the D's are dummy variables for category levels (this is like doing one hot encoding)
- one category is the reference group

I used your uploaded dataset with non-missing category and retention_label, so:

- **sample size** $n = 2737$
- **number of category levels** $= 12$
- so **number of category coefficients tested** $= 11$

For this we use logistic regression and NOT linear regression, but why?

We do not usually use ordinary linear regression here because the dependent variable retention_label is binary, not continuous. Logistic regression is appropriate because it models the probability of retention, keeps predictions between 0 and 1, and is designed for binary outcomes.

1) Defining the hypothesis

Because category is categorical and retention_label is binary, the overall logistic-regression hypothesis is:

- **Null hypothesis H_0:** category has no significant effect on retention
$$\beta_1 = \beta_2 = \cdots = \beta_{11} = 0$$
- **Alternative hypothesis H_1:** category has a significant effect on retention
$$\text{At least one } \beta_j \neq 0$$

Meaning:

- under H_0, the probability of retention is the same across categories
- under H_1, at least one category has a different retention effect

2) Deciding on the test statistic and distribution

For the **overall effect of a categorical predictor** in logistic regression, the best choice is the **Likelihood Ratio (LR) test**.

We compare:

- **null model**: no category effect
$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0$$
- **full model**: includes category
$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 D_{1i} + \cdots + \beta_{11} D_{11i}$$

Test statistic

$$LR = -2\log\, L_0 - (-2\log\, L_1)$$

equivalently,

$$LR = 2(\log\, L_1 - \log\, L_0)$$

where:

- L_0= likelihood of the null model
- L_1= likelihood of the full model

Distribution under H_0

Under the null hypothesis, the LR statistic approximately follows a **Chi-square distribution**:

$$LR \sim \chi^2_{df}$$

with

$$df = 11$$

because adding category adds 11 dummy coefficients.

Why is this test preferred here ?

A few categories in your data have very small counts and near-complete separation, so **individual Wald tests on coefficients are unstable**. The **overall LR test** is more appropriate.

3) Computing the test statistic under H_0

We will consider both mathematical and code to perform this,

Null model log-likelihood (you don't need to know how log value was derived here)

$$\log L_0 = -1203.5091$$

Full model log-likelihood (you don't need to know how log value was derived here)

$$\log L_1 = -1191.7963$$

So,

$$LR = 2(-1191.7963 - (-1203.5091))$$
$$LR = 2(11.7128)$$
$$LR = 23.4256$$

So the observed test statistic is:

$$LR = 23.4256$$

with

$$df = 11$$

and the p-value is:

$$p = 0.01538$$

Using the code we can get the sae results,

```
# 1) Load dataset
df = pd.read_csv("dimension_01_stage_01_preprocessing_part02.csv")
data = df[["category", "retention_label"]].dropna().copy()
data["retention_label"] = data["retention_label"].astype(int)

# 3) Fit the full logistic regression model -> # retention_label is binary outcome # category is categorical predictor
full_model = smf.logit("retention_label ~ C(category)",data=data).fit(disp=False)

# 4) Extract log-likelihood values
ll_full = full_model.llf       # log-likelihood of full model
ll_null = full_model.llnull    # log-likelihood of intercept-only model

# 5) Compute the LR test statistic under H0 manually
# H0: all category coefficients = 0
lr_stat = -2 * (ll_null - ll_full)

# 6) Or directly from statsmodels
lr_stat_builtin = full_model.llr
p_value = full_model.llr_pvalue

# 7) Degrees of freedom for the overall category effect
# If category has k levels, df = k - 1
df_lr = data["category"].nunique() - 1

print("Null model log-likelihood (llnull):", ll_null)
print("Full model log-likelihood (llf):", ll_full)
print("LR statistic (manual):", lr_stat)
print("Degrees of freedom:", df_lr)
print("p-value:", p_value)
```

```
Null model log-likelihood (llnull): -1203.509109489626
Full model log-likelihood (llf): -1191.7962994915692
LR statistic (manual): 23.42561999611371
Degrees of freedom: 11
p-value: 0.015384488570160483
```

4) Defining the rejection region

Take significance level:

$$\alpha = 0.05$$

For a Chi-square test with 11 degrees of freedom, reject H_0if:

$$LR > \chi^2_{0.95,11}$$

Critical value:

$$\chi^2_{0.95,11} = 19.6751$$

So the rejection region is:

$$LR > 19.6751$$

Observed value:

$$LR = 23.4256$$

Since:

$$23.4256 > 19.6751$$

the statistic falls in the rejection region.
Also, using the p-value rule:

$$p = 0.01538 < 0.05$$

So we reject H_0.

5) Conclusion

At the 5% significance level, the logistic regression shows that **category has a significant overall effect on retention_label**.
So, the conclusion is: The probability of retention is not the same across all categories.